

Networking Legacy Services and Web Standards — Detailed Lecture Notes

JKSSB Computer Awareness

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53.1 How to read this lesson

This is a trap-drill lesson. Almost every concept here has a near-neighbour: another term that sounds similar or sits next to it in a list but does a different job. The exam builds its wrong options out of exactly those neighbours, and it often asks the question with a negative stem (NOT, EXCEPT, or an incorrect statement). The defence is always the same: state what the term is, what it is not, and which neighbour it is confused with.

Term	Meaning in this lesson
Circuit switching	A communication method that reserves a dedicated path for the whole session before data moves.
Packet switching	A communication method that breaks data into packets, each travelling independently with no reserved path.
Telnet	TELEcommunication NETwork; a remote-login service for running commands on a distant computer.
FTP	File Transfer Protocol; the service for uploading and downloading files.
SMTP	Simple Mail Transfer Protocol; sends mail from the sender toward the receiver's server.
POP3	Post Office Protocol version 3; downloads mail to the local machine.
IMAP	Internet Message Access Protocol; keeps mail on the server and syncs it across devices.
W3C	World Wide Web Consortium; the body that develops open web standards.
DHTML	Dynamic HyperText Markup Language; HTML combined with styling and scripting for dynamic pages.
URL	Uniform Resource Locator; the full address of a web resource.
DNS	Domain Name System; translates domain names into IP addresses.
IP address	The numeric address of a device on the network.
OSI model	The seven-layer networking reference model.
Physical port	A hardware socket on the machine.
Logical port	A number identifying a service or process in networking software.

53.2 Packet switching vs circuit switching

Circuit switching establishes a dedicated communication path between the two ends before any data flows, and that path stays reserved for the whole session. The classic example is the old telephone call: the line is yours until you hang up, whether or not anyone is speaking.

Definition (Circuit switching). *A switching method in which a dedicated path is set up and reserved for the entire communication session before data transfer begins.*

Packet switching works the opposite way. Data is broken into small units called packets, and each packet travels independently through the network. No path is reserved in advance; packets from many senders share the same links. The Internet runs on packet switching.

Definition (Packet switching). *A switching method in which data is divided into packets that travel independently over shared links, with no dedicated path reserved in advance.*

Example 53.2.1. *An item states, “The Internet reserves a dedicated circuit for each browsing session.” This is wrong. The Internet uses packet switching: pages arrive as independent packets over shared links, and no path is reserved for the session. The wrong option has described circuit switching instead.*

53.3 Telnet and FTP

Telnet (TELEcommunication NETwork) is a remote-login service. It lets a user connect to another computer over the network and run commands there as if sitting at its keyboard. Telnet sends its traffic, including passwords, in plain text. It is not a file-transfer service and not a mail service.

FTP (File Transfer Protocol) is the file-transfer service. It moves files between computers: uploading from the local machine to a server and downloading back. It is not a remote-login shell and not a mail protocol.

The exam tests the triple together: Telnet logs in remotely, FTP moves files, and SMTP sends mail. Any option that swaps these roles — “Telnet transfers files” or “FTP sends mail” — is a planted near-neighbour distractor.

Definition (Telnet). *A network service that provides remote login to another computer so the user can run commands on it.*

Definition (FTP). *A network protocol used to transfer files between computers on a network.*

53.4 SMTP, POP3, and IMAP

Mail handling is split across three protocols, and the exam tests the split. **SMTP** (Simple Mail Transfer Protocol) *sends* mail: it carries a message from the sender’s machine toward the receiver’s mail server. **POP3** (Post Office Protocol version 3) *receives by download*: it pulls mail down to the local machine, typically removing it from the server. **IMAP** (Internet Message Access Protocol) *receives by sync*: it keeps mail on the server so the same mailbox can be read from several devices.

Definition (SMTP / POP3 / IMAP roles). *SMTP sends mail; POP3 downloads mail to the local machine; IMAP keeps mail on the server and synchronises it across devices.*

Three attached traps come from real papers. First, an option saying “POP3 keeps mail synced on the server” is wrong; that synced behaviour is IMAP (TRAP-17). Second, “a digital signature encrypts the body so only the recipient can read it” is wrong: a digital signature proves authenticity and integrity, while encryption provides confidentiality (TRAP-15). Third, “BCC hides the sender

and the recipients from each other” is wrong: BCC hides recipients from each other, the sender always knows everyone, and To and Cc names remain visible to all recipients (TRAP-16). Finally, in any mail address the @ sign separates the user name from the domain name (TRAP-28).

Example 53.4.1. *An item asks which protocol downloads mail to the local machine. The answer is POP3. If the stem instead asks which keeps the mailbox synced on the server across a phone and a desktop, the answer is IMAP, and if it asks which sends the message in the first place, the answer is SMTP.*

53.5 W3C and DHTML

The **W3C** (World Wide Web Consortium) develops the open standards of the World Wide Web, such as HTML and CSS. It is a standards body — not a browser, not a search engine, and not an Internet service provider. The near-neighbour is the **ISP** (Internet Service Provider), which sells Internet access; W3C writes the standards that the access carries.

Definition (W3C). *The World Wide Web Consortium: the international body that develops open standards for the World Wide Web.*

HTML (HyperText Markup Language) marks up the static structure of a web page. **DHTML** (Dynamic HyperText Markup Language) is not a separate replacement language; it is the combination of HTML with styling and scripting that makes pages respond and change. The trap option calls DHTML a brand-new language that replaces HTML. The correct statement is that DHTML builds on HTML to add dynamic behaviour.

53.6 URL, DNS, and IP addresses

A **URL** (Uniform Resource Locator) is the full address of a resource on the web. A **domain name** is the human-readable part inside it. The **DNS** (Domain Name System) translates that human-readable name into a numeric address. The **IP address** is that numeric address: it identifies a device on the network.

The chain in one line is: type a URL, DNS resolves its domain name, and the browser reaches the IP address. The exam swaps the links: an option saying “DNS is the numeric address” confuses DNS with the IP address, and one saying “the URL is the numeric address” confuses the URL with the IP address.

53.7 IPv4 vs IPv6

IPv4 uses 32-bit addresses, written as four decimal groups. The address space proved too small, so **IPv6** was introduced with 128-bit addresses and a fixed header of 40 bytes (PYQ-18).

Definition (IPv4 / IPv6 address size). *IPv4 addresses are 32 bits long; IPv6 addresses are 128 bits long.*

The single most repeated distractor on this topic claims IPv6 is 256-bit (TRAP-18, from JA 2026 Q78). It is always wrong: IPv6 is 128-bit, never 256-bit. A second attached trap from the same paper concerns companion protocols: **ICMP** is a control and diagnostic protocol that carries no application data and uses no ports (TRAP-19); **BGP** routes between autonomous systems (inter-domain) and does not route inside one local network (TRAP-20); and **HTTP/3** runs over QUIC over UDP, not over TCP (TRAP-21, from JA 2026 Q79).

53.8 The seven OSI layers

The **OSI model** (Open Systems Interconnection) is the seven-layer reference model for networking. From top to bottom the layers are Application, Presentation, Session, Transport, Network, Data Link, and Physical. A memory hook is “All People Seem To Need Data Processing”.

Definition (OSI model). *A seven-layer reference model for network communication: Application, Presentation, Session, Transport, Network, Data Link, Physical.*

The near-neighbour trap names four layers. Four is the layer count of the TCP/IP model, not the OSI model. When the stem says OSI, the number is always seven; an option offering four has imported the wrong model.

53.9 Physical vs logical ports

A **physical port** is a hardware socket on the machine: the place where a cable or device plugs in. A **logical port** is a number in networking software that identifies a service or process. The trap swaps them: a logical port number is not a hardware socket, and a physical socket is not a service number.

Definition (Physical vs logical port). *A physical port is a hardware socket for cables or devices; a logical port is a number identifying a network service or process.*

53.10 Exam retrieval and common confusions

Nine distinctions prevent most errors on this topic.

1. Circuit switching reserves a dedicated path; packet switching sends independent packets with no reserved path.
2. Telnet gives remote login; FTP transfers files; SMTP sends mail. Do not swap the three roles.
3. POP3 downloads mail to the local machine; IMAP keeps mail synced on the server (TRAP-17).
4. A digital signature proves authenticity and integrity; encryption provides confidentiality (TRAP-15).
5. BCC hides recipients from each other; the sender always knows all recipients (TRAP-16).
6. IPv6 is 128-bit with a 40-byte header, never 256-bit (TRAP-18).
7. ICMP carries no application data; BGP is inter-domain; HTTP/3 uses QUIC over UDP (TRAP-19, TRAP-20, TRAP-21).
8. The OSI model has seven layers; four layers is the TCP/IP model.
9. A physical port is a hardware socket; a logical port is a service number.

Example 53.10.1. *An item asks which statement is incorrect. Work each option against its near-neighbour: does the switching option reserve or share the path, does the protocol option match the service to its role, does the address option use 128-bit for IPv6, does the layer option say seven for OSI? The option that breaks its pair is the incorrect statement the stem asks for.*

Exercises

1. Distinguish circuit switching from packet switching in one sentence each.
2. Match Telnet, FTP, and SMTP to their roles.
3. State the roles of SMTP, POP3, and IMAP, and say which one keeps mail synced on the server.
4. Expand W3C, DHTML, URL, DNS, FTP, and SMTP.
5. State the bit lengths of IPv4 and IPv6, and name the OSI layers from top to bottom.

Solutions

1. Circuit switching reserves a dedicated path for the whole session before data moves; packet switching breaks data into independently travelling packets with no reserved path.
2. Telnet gives remote login to another computer; FTP transfers files between computers; SMTP sends mail toward the receiver's server.
3. SMTP sends mail, POP3 downloads it to the local machine, and IMAP keeps it on the server and syncs it; the synced-on-server protocol is IMAP.
4. W3C — World Wide Web Consortium; DHTML — Dynamic HyperText Markup Language; URL — Uniform Resource Locator; DNS — Domain Name System; FTP — File Transfer Protocol; SMTP — Simple Mail Transfer Protocol.
5. IPv4 is 32-bit and IPv6 is 128-bit; the OSI layers from the top are Application, Presentation, Session, Transport, Network, Data Link, Physical.